

# Formation of bound composite vortices of a singly-quantized $^1S_0$ vortex and half-quantized $^3P_2$ vortices in the $^1S_0$ – $^3P_2$ coexisting phase in neutron stars

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**Background:** Pulsar glitches, sudden spin-ups of rotational frequency of rotating neutron stars, have been considered to be caused by dynamics of quantized vortices of superfluid neutrons, especially  $^1S_0$  superfluid in the inner crust of neutron stars. In the outer core of neutron stars, neutrons are in the  $^3P_2$  superfluid phase, which can form half-quantized vortices (HQVs), manifesting distinct characters as compared to usual singly-quantized vortices (SQVs) of  $^1S_0$  superfluid. It has recently been suggested that the coupling between these two vortex species near the crust–core boundary gives rise to a large-scale vortex network, providing a possible explanation of the observed glitch-size distribution.

**Purpose:** We aim to elucidate the effect of interaction between a SQV in  $^1S_0$  neutron superfluid and HQVs in  $^3P_2$  neutron superfluid, which may be important near the crust–core boundary in neutron stars, where those two phases may coexist.

**Methods:** Using the Gross-Pitaevskii equations (GPE) for the  $^1S_0$  Bose-Einstein condensate (BEC) and  $^3P_2$  spinor BEC, we perform two-dimensional (2D) simulations of a single  $^1S_0$  SQV and two  $^3P_2$  HQVs in a  $^1S_0$ – $^3P_2$  coexisting phase, varying coupling constants of interaction between the two species.

**Results:** We perform 2D GPE calculations for the  $^1S_0$ – $^3P_2$  coexisting phase with varying the magnitude of the coupling constants of the density-density as well as Josephson coupling terms. From the results, we find that the Josephson coupling, arising from the relative phase between the  $^1S_0$  and  $^3P_2$  condensates, induces a strong attractive interaction between two HQVs and a single SQV, which dominates over the density-density coupling term. When pinning potentials are applied to the  $^3P_2$  HQVs and the  $^1S_0$  SQV at spatially separated locations, the Josephson-term-driven attraction is found to be sufficiently strong to overcome the pinning potential, demonstrating that the inter-condensate Josephson coupling can significantly alter the spatial configuration of quantized vortices near the  $^1S_0$ – $^3P_2$  phase boundary.

**Conclusions:** The results demonstrate that the Josephson coupling plays a crucial role in the interaction between  $^3P_2$  HQVs and a  $^1S_0$  SQV and suggest that two HQVs and a single SQV can form a bound composite vortex at the phase boundary near the inner crust and the outer core of neutron stars.

## I. INTRODUCTION

Neutron stars are among the densest objects in the universe, with central densities exceeding nuclear saturation density [1], making them a unique laboratory at the intersection of particle physics, nuclear physics, and astrophysics. Formed as compact remnants of core-collapse supernovae, neutron stars rotate rapidly, with spin periods ranging from approximately 1 ms to 10 s [2]. A subset of neutron stars emit a beam of electromagnetic radiation from their magnetic pole. Because this emission is observed on the Earth as periodic pulses synchronized with the rotation, these objects are called *pulsars* [1]. The interior of neutron stars reaches densities far exceeding those achievable in terrestrial nuclear experiments, making it the only known realization of supranuclear-density matter in the universe [1,3].

Among the observable properties of pulsars, the spin period can be measured with exceptional precision via radio or X-ray timing [2]. The neutron star mass can be measured relatively well, especially for binary systems. In contrast, the radius has remained difficult to constrain and has only recently been measured to meaningful precision, primarily through X-ray pulse-profile modeling by the Neutron Star Interior Composition Explorer (NICER) [4,5]. The magnetic field strength and its internal configuration remain subject to considerable observational uncertainty. Thermal X-ray observations of neutron star surfaces provide an additional window into the stellar interior. For instance, the rapid cooling of the Cassiopeia A neutron star, observed in real time by the *Chandra* X-ray Observatory, has been interpreted as possible direct evidence for the onset of  $^3P_2$  neutron superfluidity in the stellar core [6–8] or any other processes that cause rapid cooling such as Direct Urca processes [9]. This could be the first observational indication that the neutron  $^3P_2$  superfluidity occurs in the outer core of neutron stars (see, *e.g.*, Refs. [2,10–12] for reviews for superfluidity). Furthermore, the frictional motion of pinned superfluid vortices during spin-down—known as

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vortex creep—has been proposed as an internal heating mechanism that may explain the unexpectedly high surface temperatures of old neutron stars [13–15], connecting vortex dynamics to neutron star thermal observations.

A hallmark observational phenomenon associated with neutron stars is the *glitch* [16,17], a sudden, quasi-periodic spin-up of the pulsar rotation frequency [18,19]. The glitch mechanism is widely believed to involve dynamics of quantized vortices of superfluid neutrons. The conventional vortex avalanche model attributes glitches to the sudden unpinning of  $^1\text{S}_0$  superfluid vortices in the inner crust (see, e.g., Refs. [13,20–31], and references therein). However, a growing number of observations have revealed glitch events that cannot be explained solely by this model [32,33], calling for a new theoretical framework that can describe the full diversity of observed glitch phenomena.

In the outer core of neutron stars, neutrons are expected to be in the  $^3\text{P}_2$  superfluid phase [34–51], forming a spin-triplet  $p$ -wave condensate that is realized on the Earth in only a handful of systems such as superfluid  $^3\text{He}$ . Moreover, the outer core allows coexistence of  $^3\text{P}_2$  neutron superfluid and  $^1\text{S}_0$  proton superconductor, offering rare environment for a multi-component condensate system of broad interest. On the theoretical side, significant progress has been made in the microscopic understanding of the  $^3\text{P}_2$  superfluid. The Ginzburg-Landau (GL) framework [51–55] enables us to study the ground states as well as nonuniform states such as quantized vortices. The ground state is a nematic phase [52], and more precisely, it is in the uniaxial (UN),  $\text{D}_2$  biaxial nematic ( $\text{D}_2\text{-BN}$ ), or  $\text{D}_4$  biaxial nematic ( $\text{D}_4\text{-BN}$ ) phases depending on the strength of magnetic field and temperature [53–55]. Quantized vortices in the  $^3\text{P}_2$  superfluid have been studied in the GL framework [53,56–61]. Internal degrees of freedom of the  $^3\text{P}_2$  superfluid allow for presence of half-quantized vortices (HQVs) in strong magnetic field in which the system is in the  $\text{D}_4\text{-BN}$  phase [58]. These vortices are also characterized by the non-Abelian fundamental group and are called non-Abelian HQVs. A singly quantized vortex (SQV) splits into two HQVs, forming a vortex molecule structure [60], and the interaction between two or more molecules was studied [61]. Other than vortices, domain walls [62] and surface defects [63] were also studied. There are also rich massless and massive bosonic excitations [64–76], which were proposed to explain cooling and transport phenomena in neutron stars.

Moreover, recently the  $^3\text{P}_2$  superfluid has been studied microscopically by using the Eilenberger and Bogoliubov-de Gennes (BdG) equations [77–79] which are applicable near the transition temperature and can describe short distance behaviors such as vortex cores. The complete phase diagram under strong magnetic field and temperature as well as topological superfluidity such as topologically-protected gapless Majorana fermions on surfaces and vortex cores were shown. Subsequently, the first microscopic study of quantized vortices in the  $^3\text{P}_2$  superfluid was made in the BdG equation [80–82]. In particular, a SQV was found to host two zero-energy Majorana fermion bound states in its core [80]. Such a SQV splits into two non-Abelian HQVs [81], consistent with the analysis in the GL theory, and each HQV hosts a single zero-

energy Majorana fermion bound state.

Separately, the coexistence of  $^1\text{S}_0$  and  $^3\text{P}_2$  superfluids were studied within the GL framework [83], obtaining the phase diagram as a function of temperature and magnetic field. In Ref. [83], it was shown that the coupling between the condensates for the  $^1\text{S}_0$  and  $^3\text{P}_2$  superfluids is a form of the *Josephson coupling* in the quadratic order in terms of both the condensations. Further, it was found that the two condensates can coexist under weak magnetic fields, and the parameter region of the  $\text{D}_4\text{-BN}$  phase is enhanced to the whole the  $^3\text{P}_2$  superfluid region. These results indicate that the internal structure of  $^3\text{P}_2$  vortices and the thermodynamic properties of the coexistence phase are strongly influenced by the interplay between the two superfluid components. However, a full microscopic simulation of dynamics of  $^3\text{P}_2$  superfluid vortices in the coexistence phase has not been carried out yet.

In the outer core, the  $^3\text{P}_2$  superfluid phase hosts vortices with qualitatively different properties from those in the inner crust, including the possibility of HQVs [61]. Building on this distinction, it was proposed in Ref. [84] that the coupling between outer-core  $^3\text{P}_2$  vortices and inner-crust  $^1\text{S}_0$  vortices of superfluid neutrons gives rise to a large-scale vortex network, which has been advocated to be a candidate mechanism that explains the pulsar glitch phenomenon. Namely, the vortex network model provides a scaling law of the glitch energy that can explain the same law from the observation without fine tuning parameters. Despite this proposal, a microscopic analysis of structure and dynamics of  $^3\text{P}_2$  vortices, as well as their interaction with proton superconducting flux tubes in the outer core, has not yet been carried out [85]. In particular, if the  $^3\text{P}_2$  and  $^1\text{S}_0$  superfluids are overlapped in a certain region of neutron stars, vortices in these superfluids would meet and interact, forming junctions called *boojums*. These are key ingredient for a large-scale vortex network mentioned above. Josephson currents across the  $^3\text{P}_2$ – $^1\text{S}_0$  boundary were also studied [86]. The time-dependent oscillating Josephson current induced by vortex motion produces electromagnetic radiation whose power far exceeds Ohmic dissipation, potentially contributing to neutron star heating during the photon cooling era. We mention here that a one-dimensional system of coexisting s-wave and p-wave superfluids was discussed in the context of ultracold atomic gases [87].

In this work, we investigate the microscopic interaction between  $^3\text{P}_2$  and  $^1\text{S}_0$  superfluid vortices in a  $^3\text{P}_2$ – $^1\text{S}_0$  coexisting phase, by extending the thermodynamic analysis of Ref. [83] to include spatially inhomogeneous vortex configurations. We perform two-dimensional (2D) Gross-Pitaevskii (GP) calculations, which are valid at zero temperature, treating coefficients of interactions free parameters, in contrast to the GL formalism valid near the transition temperature [88]. Building upon the microscopic vortex structure identified in Ref. [53], we evaluate how the coupling between these two vortex species affects the vortex configurations, with the aim to provide a quantitative basis for understanding the diversity of observed glitch phenomena beyond the reach of the conventional vortex avalanche model that relies only on the  $^1\text{S}_0$  superfluid in the inner crust of neutron stars.

Vortex molecule structure in multi-component condensates

has attracted substantial interest in condensed matter physics. For instance, two HQVs are connected by a domain wall in two-component Bose-Einstein condensates (BECs) [89–97], two-gap or two-component superconductors [98–101], and chiral  $p$ -wave superconductors [102,103]. Such configurations were extended to more complicated vortex molecules in multi-components systems [104–109] and/or arbitrary charges [110,111]. In this work, we show that a vortex molecule of a form of HQV–SQV–HQV can be formed near the crust–core boundary, which consists of a single  $^1\text{S}_0$  SQV and two  $^3\text{P}_2$  HQVs which are connected to the SQV by a domain wall due to the quadratic Josephson coupling.

The article is organized as follows. In Sec. II, we introduce the GP framework used to describe coexisting  $^3\text{P}_2$  and  $^1\text{S}_0$  superfluids, including  $^3\text{P}_2$  and  $^1\text{S}_0$  interaction. In Sec. III, we outline the numerical procedure for various vortex configurations. In Sec. IV, we present the results of GP calculations and discuss how vortex configurations change depending on the coupling constants for the interaction terms. Finally, a summary and prospect are given in Sec. V.

## II. METHODS

### A. Total free energy functional

We consider the coexisting phase of  $^1\text{S}_0$  and  $^3\text{P}_2$  neutron superfluids, which may be realized near the crust-core boundary in neutron stars. To describe such a coexisting phase of  $^1\text{S}_0$  and  $^3\text{P}_2$  neutron superfluids, we employ the GP equations. In this article, we denote the complex scalar order parameter of the  $^1\text{S}_0$  superfluid as  $\sigma(\mathbf{r})$ , while the five-component complex spinor order parameter of the  $^3\text{P}_2$  superfluid is denoted as  $\vec{\psi}(\mathbf{r}) = (\psi_{-2}(\mathbf{r}), \psi_{-1}(\mathbf{r}), \psi_0(\mathbf{r}), \psi_1(\mathbf{r}), \psi_2(\mathbf{r}))^T$ . We restrict our numerical analyses to 2D coordinate space, while the equations will be given in a general form that can be applied to three-dimensional (3D) systems.

The total free energy functional of the mixed system of  $^1\text{S}_0$  and  $^3\text{P}_2$  superfluids is given by

$$F_{\text{total}}[\sigma, \vec{\psi}] = F_{\text{singlet}}[\sigma] + F_{\text{triplet}}[\vec{\psi}] + F_{\text{int}}[\sigma, \vec{\psi}], \quad (1)$$

where  $F_{\text{singlet}}[\sigma]$  and  $F_{\text{triplet}}[\vec{\psi}]$  are the free energy functionals of the  $^1\text{S}_0$  and  $^3\text{P}_2$  condensates, respectively, and  $F_{\text{int}}[\sigma, \vec{\psi}]$  accounts for contributions from the inter-condensate interaction. Each term is discussed in detail in the following subsections. The equilibrium configurations of  $\sigma(\mathbf{r})$  and  $\vec{\psi}(\mathbf{r})$  are obtained by minimizing  $F_{\text{total}}[\sigma, \vec{\psi}]$  with respect to the order parameters, yielding the coupled Euler-Lagrange equations:

$$\frac{\delta F_{\text{total}}}{\delta \sigma^*} = 0, \quad (2)$$

$$\frac{\delta F_{\text{total}}}{\delta \psi_m^*} = 0 \quad (m = -2, \dots, 2). \quad (3)$$

In practice, we solve these equations numerically using the imaginary-time method, as described in Sec. III D.

### B. Free energy functional of the $^1\text{S}_0$ neutron superfluid

The  $^1\text{S}_0$  neutron superfluid is characterized by a complex scalar order parameter  $\sigma(\mathbf{r})$ , which represents the condensate of spin-singlet Cooper pairs. Within the GP framework, the free energy functional is given by

$$F_{\text{singlet}}[\sigma] = \int \left[ \frac{\hbar^2}{2M} |\nabla \sigma(\mathbf{r})|^2 + U_{\text{trap}}(\mathbf{r}) |\sigma(\mathbf{r})|^2 + \frac{g_0}{2} |\sigma(\mathbf{r})|^4 \right] d\mathbf{r}, \quad (4)$$

where  $M = 2m_n$  is the mass of a neutron Cooper pair, being  $m_n$  the neutron mass, and  $g_0$  is a constant that controls the Landau potential. The first term is the gradient energy, which penalizes spatial variations of the order parameter and governs the coherence length of the condensate. The second term is the trapping potential that confines the condensates in a finite region. The third term describes the interaction that governs non-linear effects.

### C. Free energy functional of the $^3\text{P}_2$ neutron superfluid

Since the total angular momentum of a  $^3\text{P}_2$  Cooper pair is  $J = 2$ , the  $^3\text{P}_2$  neutron superfluid can be formally modeled as a spin-2 spinor BEC. We therefore adopt the spin-2 GP framework for spinor BEC [112,113] to describe the  $^3\text{P}_2$  condensate. Among possible phases of the spin-2 spinor BECs, the relevant one for the  $^3\text{P}_2$  neutron superfluid is the nematic phase [114,115]. The free energy functional of the  $^3\text{P}_2$  superfluid reads [112]

$$F_{\text{triplet}}[\vec{\psi}] = \int \left\{ \sum_{m=-2}^2 \psi_m^*(\mathbf{r}) \left[ -\frac{\hbar^2}{2M} \nabla^2 + U_{\text{trap}}(\mathbf{r}) \right] \psi_m(\mathbf{r}) + \frac{c_0}{2} n^2(\mathbf{r}) + \frac{c_1}{2} |\mathbf{F}(\mathbf{r})|^2 + \frac{c_2}{10} |\Psi_{20}(\mathbf{r})|^2 \right\} d\mathbf{r}, \quad (5)$$

where  $n(\mathbf{r})$  is the superfluid density of the  $^3\text{P}_2$  superfluid,  $n(\mathbf{r}) = \sum_{m=-2}^2 |\psi_m(\mathbf{r})|^2$ ,  $\mathbf{F}(\mathbf{r})$  is the spin-density vector, and  $\Psi_{20}(\mathbf{r})$  is a gauge-invariant scalar quantity characterizing the condensate.  $c_i$  ( $i = \{0, 1, 2\}$ ) are coupling constants that characterize the density-density, spin-spin, and spin-singlet pair interactions, respectively.

Each component of the spin-density vector  $\mathbf{F}(\mathbf{r})$  is defined by

$$F_v(\mathbf{r}) = \sum_{m,m'=-2}^2 \psi_m^*(\mathbf{r}) (f_v)_{mm'} \psi_{m'}(\mathbf{r}), \quad (6)$$

where  $f_v$  ( $v = \{x, y, z\}$ ) are the standard  $5 \times 5$  spin matrices for

spin-2, whose explicit forms can be found in, *e.g.*, Ref. [112]:

$$f_x = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & \sqrt{\frac{3}{2}} & 0 & 0 \\ 0 & \sqrt{\frac{3}{2}} & 0 & \sqrt{\frac{3}{2}} & 0 \\ 0 & 0 & \sqrt{\frac{3}{2}} & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \quad (7)$$

$$f_y = \begin{pmatrix} 0 & -i & 0 & 0 & 0 \\ i & 0 & -\sqrt{\frac{3}{2}}i & 0 & 0 \\ 0 & \sqrt{\frac{3}{2}}i & 0 & -\sqrt{\frac{3}{2}}i & 0 \\ 0 & 0 & \sqrt{\frac{3}{2}}i & 0 & -1 \\ 0 & 0 & 0 & i & 0 \end{pmatrix}, \quad (8)$$

$$f_z = \begin{pmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -2 \end{pmatrix}. \quad (9)$$

The  $^3P_2$  superfluid order parameter is often represented using a  $3 \times 3$  complex symmetric matrix  $A(\mathbf{r})$ , having matrix elements  $A_{ij}(\mathbf{r})$  ( $i, j = \{x, y, z\}$ ), which encodes the spin-triplet,  $p$ -wave pairing structure of the condensate. The matrix element  $A_{ij}(\mathbf{r})$  can be decomposed in terms of the magnetic quantum number components  $\psi_m(\mathbf{r})$  ( $m = -2, \dots, 2$ ) as [116]

$$A_{ij}(\mathbf{r}) = \sum_{m=-2}^2 \psi_m(\mathbf{r}) \left( \mathcal{Y}_m^{(2)} \right)_{ij}, \quad (10)$$

where  $\mathcal{Y}_m^{(2)}$  are the rank-2 symmetric traceless tensors constructed from the spin-2 spherical harmonics, forming a basis for the irreducible representation of the rotation group  $SO(3)$  in the spin-2 sector [116].

The gauge-invariant scalar quantity characterizing the condensate,  $\Psi_{20}(\mathbf{r})$ , is constructed from the trace of  $A^2(\mathbf{r})$  which is invariant under  $SO(3)$  spin rotations, but transforms non-trivially under the  $U(1)$  gauge transformation  $A_{ij} \rightarrow e^{2i\alpha} A_{ij}$ , acquiring a phase  $e^{4i\alpha}$ . Using the orthogonality of the basis tensors  $\mathcal{Y}_m^{(2)}$ , one obtains

$$\Psi_{20}(\mathbf{r}) \equiv \text{tr}[A^2(\mathbf{r})] = \sum_{m=-2}^2 (-1)^m \psi_m(\mathbf{r}) \psi_{-m}(\mathbf{r}), \quad (11)$$

where the factor  $(-1)^m$  arises from the contraction of the basis tensors, reflecting the time-reversal properties of the spin-2 representation [116].

The quantity  $\Psi_{20}$  plays a central role in identifying the positions of HQVs in the  $^3P_2$  condensate. Under a  $2\pi$  rotation around a HQV, the order parameter  $\psi_m$  acquires a phase of  $e^{i\pi} = -1$ , leaving the physical state unchanged due to the discrete  $\mathbb{Z}_2$  symmetry of the order parameter manifold [61]. Consequently,  $\Psi_{20}$ , being quadratic in  $\psi_m$ , acquires a phase of  $e^{2i\pi} = 1$  under a single circuit around a HQV, whereas it winds by  $e^{4i\pi}$  around an SQV. This distinction makes  $\Psi_{20}$

a sensitive probe of the vortex topology: the phase singularities of  $\Psi_{20}(\mathbf{r})$ —points around which its phase winds by  $2\pi$ —correspond to the cores of individual HQVs, as opposed to the integer vortices of  $\sigma(\mathbf{r})$ . The vortex core positions of the HQVs are therefore identified numerically by locating the phase singularities of  $\Psi_{20}(\mathbf{r})$  on the simulation lattice.

#### D. Free energy functional of inter-condensate interaction

In the coexistence region near the crust-core boundary, both the  $^1S_0$  and  $^3P_2$  superfluid phases may be present simultaneously. In such a  $^1S_0$ - $^3P_2$  coexisting region, the two order parameters  $\tilde{\psi}(\mathbf{r})$  and  $\sigma(\mathbf{r})$  are coupled through an interaction term in the free energy functional. Following the GL framework, and adopting the structure of the coupling term introduced in Ref. [83], the interaction free energy between the two condensates is composed of two terms:

$$F_{\text{int}}[\tilde{\psi}, \sigma] = \int \left[ \mathcal{E}_{\text{den}}(\mathbf{r}) + \mathcal{E}_{\text{Jos}}(\mathbf{r}) \right] d\mathbf{r} \quad (12)$$

where  $\mathcal{E}_{\text{den}}(\mathbf{r})$  and  $\mathcal{E}_{\text{Jos}}(\mathbf{r})$  represent the interaction energy densities associated with the density-density and the Josephson couplings, respectively. They are given by

$$\begin{aligned} \mathcal{E}_{\text{den}}(\mathbf{r}) &= \zeta_1 |\sigma(\mathbf{r})|^2 \text{tr}[A^*(\mathbf{r})A(\mathbf{r})] \\ &= \zeta_1 |\sigma(\mathbf{r})|^2 \left[ \sum_{m=-2}^2 (-1)^m |\psi_m(\mathbf{r})|^2 \right], \end{aligned} \quad (13)$$

$$\begin{aligned} \mathcal{E}_{\text{Jos}}(\mathbf{r}) &= \zeta_2 (\sigma^2 \text{tr}[A^{*2}(\mathbf{r})] + \sigma^{*2} \text{tr}[A^2(\mathbf{r})]) \\ &= \zeta_2 \sum_{m=-2}^2 (-1)^m [\sigma^2(\mathbf{r}) \psi_m^{*2}(\mathbf{r}) + \sigma^{*2}(\mathbf{r}) \psi_m^2(\mathbf{r})] \\ &= \zeta_2 \sum_{m=-2}^2 (-1)^m |\sigma(\mathbf{r})|^2 |\psi_m(\mathbf{r})|^2 \cos[2\Delta\theta_m(\mathbf{r})], \end{aligned} \quad (14)$$

where  $\Delta\theta_m(\mathbf{r}) \equiv \theta_\sigma(\mathbf{r}) - \theta_m(\mathbf{r})$  denotes the phase difference between the  $^1S_0$  and  $^3P_2$  order parameters, where  $\theta_\sigma(\mathbf{r}) = \arg(\sigma(\mathbf{r}))$  and  $\theta_m(\mathbf{r}) = \arg(\psi_m(\mathbf{r}))$  denote their phase fields, respectively.  $\zeta_1$  and  $\zeta_2$  are real coupling constants defined as

$$\zeta_{1,2} = \tilde{\zeta}_{1,2} \frac{4\pi\hbar^2}{M}, \quad (15)$$

with parameters  $\tilde{\zeta}_{1,2}$ . This normalization is adopted so that  $\zeta_{1,2}$  carry the same dimensions as the coupling constant  $g_0$  in Eqs. (4) and (5), allowing for a direct comparison of the interaction strength between the two condensates. The dimensionless values  $\tilde{\zeta}_{1,2}$  are varied independently in the present numerical analysis.

The total interaction energy associated with the density-density coupling,  $E_{\text{den}} = \int \mathcal{E}_{\text{den}}(\mathbf{r}) d\mathbf{r}$ , is invariant under independent  $U(1)$  phase rotations of each order parameter. The total interaction energy associated with the Josephson coupling,  $E_{\text{Jos}} = \int \mathcal{E}_{\text{Jos}}(\mathbf{r}) d\mathbf{r}$ , represents a coherent pair-exchange interaction between the two superfluid components, which breaks the relative  $U(1)$  phase symmetry between  $\tilde{\psi}(\mathbf{r})$  and  $\sigma(\mathbf{r})$ .

This type of coupling is analogous to the Josephson interaction known in coupled superconductors [111,117], and plays a crucial role in determining the structure and dynamics of quantized vortices near the phase boundary between the inner crust and the outer core. This coupling is analogous to the Rabi coupling in the context of BECs, in which it is linear with respect to both the condensates. In this work, we refer to this coupling as the Josephson coupling and associated energy  $E_{\text{Jos}}$  as Josephson energy. The relative magnitude and signs of  $\zeta_1$  and  $\zeta_2$  control whether the interaction is attractive or repulsive between the two condensates, and are determined from microscopic considerations in the present analysis.

### III. COMPUTATIONAL DETAILS

#### A. Trapping potential and discretization

Numerical simulations are performed on a 2D square lattice, where all fields are represented on a uniform mesh. In this work, we use  $100 \times 100$  grid points with a 0.1-fm mesh. To compute Laplacian of spatial functions, we employ the spectral method executing fast Fourier transform (FFT) routines of the FFTW library. Because of the use of FFT, the periodic boundary conditions are automatically adopted. To avoid vortices to interact due to the periodicity of the computational box, we confine the condensate within a trapping potential  $U_{\text{trap}}(\mathbf{r})$ , which is defined as a function of the radial distance  $r = |\mathbf{r}|$  from the center of the simulation box. The potential is characterized by a trap radius  $R_{\text{trap}}$  and a smoothing interval  $\delta$ , and is given by

$$U_{\text{trap}}(r) = \begin{cases} 0 & r < R_{\text{trap}} - \delta, \\ V \left( \frac{1}{2} - \frac{1}{2} \tanh \left[ \pi \tan \left( \frac{\pi}{2\delta} (r - R_{\text{trap}}) \right) \right] \right)^{-1} & R_{\text{trap}} - \delta \leq r \leq R_{\text{trap}} + \delta, \\ V & r > R_{\text{trap}} + \delta, \end{cases} \quad (16)$$

where  $V$  is a large positive constant that effectively excludes the condensate from the region outside the trap. The intermediate region,  $R_{\text{trap}} - \delta \leq r \leq R_{\text{trap}} + \delta$ , connects smoothly the flat interior and the hard wall, suppressing numerical artifacts arising from an abrupt potential change. In the present work, we use  $V = 10^4$  MeV,  $R_{\text{trap}} = 4.5$  fm, and  $\delta = 0.4$  fm for all the simulations given below.

#### B. Pinning potential

To independently control the positions of the  $^1\text{S}_0$  SQV and the  $^3\text{P}_2$  HQVs, we introduce pinning potentials for each condensate separately. For the  $^1\text{S}_0$  condensate, we introduce a

pinning potential as

$$F_{\text{pin,singlet}} = \int U_{\text{pin,singlet}}(\mathbf{r}) |\sigma(\mathbf{r})|^2 d\mathbf{r}, \quad (17)$$

which pins the SQV core of the  $^1\text{S}_0$  condensate at a separate location. For the  $^3\text{P}_2$  condensate, a pinning potential is coupled to the gauge-invariant density  $|\Psi_{20}(\mathbf{r})|^2$ , adding the following term to the free energy functional:

$$F_{\text{pin,triplet}} = \int U_{\text{pin,triplet}}(\mathbf{r}) |\Psi_{20}(\mathbf{r})|^2 d\mathbf{r}. \quad (18)$$

Since  $\Psi_{20}$  vanishes at the core of each HQV [see Eq. (11)], this term energetically favors the suppression of  $|\Psi_{20}|^2$  at the potential center, thereby pinning the HQV cores at prescribed locations. The GP equations are supplemented with the following terms associated with the pinning potentials:

$$\frac{\delta F_{\text{pin,singlet}}}{\delta \sigma^*} = U_{\text{pin,singlet}}(\mathbf{r}) \sigma(\mathbf{r}), \quad (19)$$

$$\frac{\delta F_{\text{pin,triplet}}}{\delta \psi_m^*} = (-1)^m U_{\text{pin,triplet}}(\mathbf{r}) \psi_{-m}^*(\mathbf{r}) \Psi_{20}(\mathbf{r}). \quad (20)$$

For the pinning potential, we use a simple form defined as

$$U_{\text{pin,singlet}}(\mathbf{r}) = \begin{cases} V & (r \leq R_{\text{pin}}) \\ 0 & (r > R_{\text{pin}}) \end{cases}, \quad (21)$$

$$U_{\text{pin,triplet}}(\mathbf{r}) = \begin{cases} V & (|\mathbf{r} \pm d_{\text{pin}} \hat{\mathbf{x}}| \leq R_{\text{pin}}) \\ 0 & \text{elsewhere} \end{cases}. \quad (22)$$

That is, each pinning site is modeled as a repulsive potential with a small spatial extent, where  $R_{\text{pin}} = 0.3$  fm is used in the present study.

#### C. Vortex initialization

To obtain solutions with quantized vortices, we introduce the desired phase winding to the initial order parameters by hand. Specifically, a phase profile corresponding to the target vortex number is assigned to  $\tilde{\psi}$  and  $\sigma$  as initial conditions, *e.g.*, in the case of the  $^1\text{S}_0$  condensate,

$$\sigma(\mathbf{r}) = |\sigma_0(\mathbf{r})| \exp[iN_v \theta_\sigma(\mathbf{r})], \quad (23)$$

where  $\sigma_0$  is the initial guess of the order parameter and  $\theta_\sigma(\mathbf{r}) = \arctan[(y - y_0)/(x - x_0)]$  is its phase, rotating about the center of the box  $(x_0, y_0)$ . In the analyses presented in this paper, the winding number is always set to  $N_v = 1$ . We adopt the same procedure also for all the components of the  $^3\text{P}_2$  condensate  $\psi_m(\mathbf{r})$  ( $m = -2, \dots, 2$ ). As the initial guess of the order parameters, we use a uniform distribution within the trap radius  $R_{\text{trap}}$ . Starting with the initial configuration, the system is evolved in imaginary time to relax toward the energetically favored vortex configuration, automatically adjusting the amplitude and the phase. This procedure yields the equilibrium spatial arrangement and number of vortices for a given set of parameters.

### D. Imaginary-time evolution

The GP equations are obtained by taking the functional derivatives of the total free energy functional, supplemented with the contributions of the pinning potentials, with respect to the order parameters:

$$i\hbar \frac{\partial \sigma(\mathbf{r})}{\partial t} = \frac{\delta F_{\text{total}}}{\delta \sigma^*(\mathbf{r})}, \quad (24)$$

$$i\hbar \frac{\partial \psi_m(\mathbf{r})}{\partial t} = \frac{\delta F_{\text{total}}}{\delta \psi_m^*(\mathbf{r})}, \quad (25)$$

yielding the coupled equations for  $\sigma(\mathbf{r})$  and  $\psi_m(\mathbf{r})$  [112]. Equilibrium vortex configurations are obtained by evolving Eqs. (24) and (25) in imaginary time ( $t \rightarrow -i\tau$ ).

Specifically, in the imaginary time evolution, we solve the following equations:

$$\begin{aligned} -\hbar \frac{\partial \sigma(\mathbf{r})}{\partial \tau} = & \left[ -\frac{\hbar^2}{2M} \nabla^2 + U_{\text{trap}}(\mathbf{r}) + g_0 |\sigma(\mathbf{r})|^2 \right] \sigma(\mathbf{r}) \\ & + \sum_{m=-2}^2 [\zeta_1 |\psi_m(\mathbf{r})|^2 \sigma(\mathbf{r}) + 2(-1)^m \zeta_2 \sigma^*(\mathbf{r}) \psi_m(\mathbf{r})] \\ & + U_{\text{pin,singlet}}(\mathbf{r}) \sigma(\mathbf{r}), \end{aligned} \quad (26)$$

$$\begin{aligned} -\hbar \frac{\partial \psi_m(\mathbf{r})}{\partial \tau} = & \left[ -\frac{\hbar^2}{2M} \nabla^2 + U_{\text{trap}}(\mathbf{r}) + c_0 n(\mathbf{r}) \right] \psi_m(\mathbf{r}) \\ & + \sum_{m'=-2}^2 c_1 \mathbf{F}^*(\mathbf{r}) \cdot (\mathbf{f})_{mm'} \psi_{m'}(\mathbf{r}) + c_2 \frac{1}{5} \psi_m^*(\mathbf{r}) \Psi_{20}(\mathbf{r}) \\ & + \zeta_1 |\sigma(\mathbf{r})|^2 \psi_m(\mathbf{r}) + 2(-1)^m \zeta_2 \psi_m^*(\mathbf{r}) \sigma^2(\mathbf{r}) \\ & + (-1)^m U_{\text{pin,triplet}}(\mathbf{r}) \psi_{-m}^*(\mathbf{r}) \Psi_{20}(\mathbf{r}), \end{aligned} \quad (27)$$

where  $\mathbf{f} = (f_x, f_y, f_z)$  and  $\tau$  represents the imaginary time.

At each step of the imaginary-time evolution, the order parameters are normalized so that the integrated particle numbers of the  $^1\text{S}_0$  and  $^3\text{P}_2$  condensates are separately conserved. Specifically, after each step,  $\sigma(\mathbf{r})$  and  $\tilde{\psi}(\mathbf{r})$  are normalized such that

$$\int |\sigma(\mathbf{r})|^2 d\mathbf{r} = N, \quad \int \sum_{m=-2}^2 |\psi_m(\mathbf{r})|^2 d\mathbf{r} = N, \quad (28)$$

where the particle number  $N$  is held fixed and equal for both condensates throughout the evolution. This normalization condition ensures that the imaginary-time evolution converges to the ground state at the fixed particle number for each component, rather than allowing the condensate densities to change freely.

### E. Vortex pinning and activation of the interaction term

To analyze the change of vortex configurations triggered by the  $^1\text{S}_0$ - $^3\text{P}_2$  coupling, we adopt the following two-stage protocol. In the first stage, the coupling constants are set to

$\zeta_1 = \zeta_2 = 0$ , completely decoupling the two superfluid components. In this decoupled state, artificial pinning potentials are introduced for both the singlet and triplet order parameters independently. The imaginary-time evolution is performed for the first 100 fm/c starting from the initial state with phase winding at  $r = 0$ . Through the imaginary-time evolution, the vortices gradually move and are eventually pinned to the pinning sites.

In the second stage, we switch on the interaction terms  $\zeta_1$  and  $\zeta_2$  at  $\tau = 100$  fm/c, activating the coupling between the  $^1\text{S}_0$  and  $^3\text{P}_2$  condensates. This protocol offers a way to numerically obtain the equilibrium configuration with the  $^1\text{S}_0$ - $^3\text{P}_2$  couplings, starting from the well-converged solution for the case without the coupling term.

### F. Vortex position detection

A quantum vortex is characterized by a phase singularity of the order parameter, near which the density of the order parameter vanishes. In the present calculations, the positions of quantum vortices are extracted by examining the phase of the order parameter.

As an example, the position of a vortex in the  $^1\text{S}_0$  order parameter  $\sigma$  is determined as follows. At each lattice point  $(i_x, i_y)$ , we collect the phase  $\arg(\sigma) = \text{atan2}(\text{Re}(\sigma), \text{Im}(\sigma))$  at the surrounding eight points  $\{(i_x - 1, i_y - 1), (i_x, i_y - 1), (i_x + 1, i_y - 1), (i_x + 1, i_y), (i_x + 1, i_y + 1), (i_x, i_y + 1), (i_x - 1, i_y + 1), (i_x - 1, i_y)\}$ . The phase differences  $\delta\theta$  between adjacent points along this loop are then summed. When the absolute value of a phase difference satisfies  $|\delta\theta| > \pi$ , the value is corrected to lie within  $[-\pi, \pi]$ , accounting for the branch of  $\text{atan2}$  whose range is  $(-\pi, \pi]$ , so that the phase is treated as a multivalued function. After applying this correction, the phase differences at the eight surrounding points are summed. We confirm that this sum is almost zero if the eight spatial points do not include the singular point. When the summed value is nearly  $2\pi$ , we regard that a vortex is present at the point  $(i_x, i_y)$ . The same procedure is applied to the  $^3\text{P}_2$  order parameter  $\Psi_{20}$ . Since  $\Psi_{20} \propto (\tilde{\psi})^2$ , the total phase winding around a single HQV represented  $\Psi_{20}$  is  $2\pi$ .

Since a single quantum of vorticity  $\kappa = 2\pi$  gives rise to two HQVs, the distance between the two HQVs serves as a useful measure for analyzing the interaction between the SQV and the HQVs. This inter-HQV distance is computed by determining the coordinates of each HQV and evaluating the Euclidean distance between them. Note that since the vortex detection is based on the phase winding integral around eight surrounding lattice points, the spatial resolution of the inter-vortex distance is fundamentally limited by the lattice spacing of the discretized computational domain.

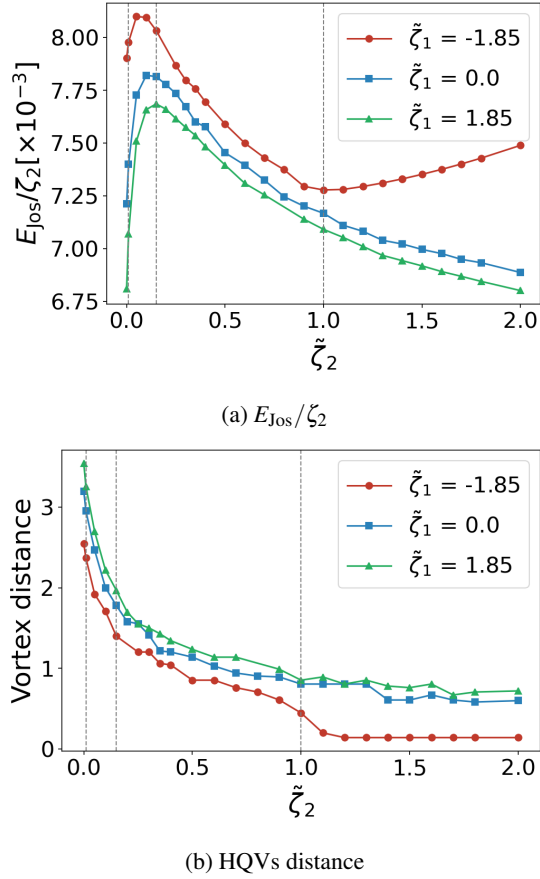


FIG. 1: (a) Josephson energy,  $E_{\text{Jos}}/\zeta_2$ , as a function of the Josephson coupling constant  $\tilde{\zeta}_2$  for three values of the density-density coupling constant  $\tilde{\zeta}_1$ , as indicated in the legend. The system consists of one SQV in the  $^1\text{S}_0$  condensate and two HQVs in the  $^3\text{P}_2$  condensate. (b) Inter-vortex distance between the two HQVs in the  $^3\text{P}_2$  condensate as a function of the Josephson coupling constant  $\tilde{\zeta}_2$  for three values of the density-density coupling constant  $\tilde{\zeta}_1$ , as indicated in the legend. The system configuration and imaginary-time evolution procedure are the same as in Fig. 1a.

## IV. RESULTS AND DISCUSSION

### A. Simulations without pinning sites

In this section, we analyze how the  $^1\text{S}_0$ - $^3\text{P}_2$  interaction affects the vortex configuration. To this end, we calculate the vortex configurations for various  $\zeta_1$  and  $\zeta_2$  values, without introducing pinning site.

#### 1. Energy and Vortex Distance

In Figs. 1(a) and 1(b), we show the Josephson energy,  $E_{\text{Jos}}/\zeta_2$ , and the distance between two HQVs in the  $^3\text{P}_2$  condensate, respectively, as a function of the coupling constant

$\tilde{\zeta}_2$ , for three representative values of  $\tilde{\zeta}_1$ . Results for  $\tilde{\zeta}_1 = -1.85$  (attractive density-density coupling), 0 (no coupling), and 1.85 (repulsive density-density coupling) are shown by red filled circles, blue filled squares, and green filled triangles, respectively, connected with solid lines. Since  $E_{\text{Jos}}$  is linear in  $\zeta_2$  by construction [see Eq. (14)], plotting  $E_{\text{Jos}}/\zeta_2$  isolates the geometric contribution arising purely from the phase structure of the vortices, independently of the coupling strength.

Even for the  $\tilde{\zeta}_2 = 0$  case, the Josephson energy shows a small, but finite value, because  $^1\text{S}_0$  and  $^3\text{P}_2$  condensate are independent from each other and, thus, there exists, in general, finite phase difference among them. When the Josephson coupling is introduced ( $0 < \tilde{\zeta}_2 \lesssim 0.1$ ),  $E_{\text{Jos}}/\zeta_2$  increases rapidly, developing characteristic phase structure in the  $^1\text{S}_0$  and  $^3\text{P}_2$  condensates. As the  $\tilde{\zeta}_2$  increases further, the Josephson energy decreases with decreasing vortex distance, see Fig. 1(b). That is, since the Josephson coupling costs additional energy associated with the phase difference, the system tends to reduce the distance between two  $^3\text{P}_2$  HQVs, as a result of energy balance between gain with reducing  $E_{\text{Jos}}$  and loss associated with the repulsive interaction between two HQVs. This trend continues up to  $\tilde{\zeta}_2 = 2$  for the  $\tilde{\zeta}_1 = 0$  and 1.85 cases.

Figure 1b also indicates that the sign of  $\zeta_1$  affects the vortex distance. The positive (negative)  $\zeta_1$  increases (decreases) the distance compared with the case of  $\zeta_1 = 0$ . This implies that positive (negative)  $\zeta_1$  yields repulsion (attraction) between the vortices. This is consistent with the case of two-component BECs [118]. In particular, when  $\zeta_1$  is attractive, for  $\tilde{\zeta}_2 \gtrsim 1$ , the vortex distance decreases quickly and eventually reaches a small value of 0.1–0.2 fm. It is because of the fact that  $\tilde{\zeta}_1 < 0$  corresponds to attraction for a larger density overlap, thereby the system prefers to place  $^1\text{S}_0$  and  $^3\text{P}_2$  vortices as close as possible. We note that since we work with mesh spacing of  $\Delta x = 0.1$  fm, we can not resolve vortex distance shorter than  $\mathcal{O}(\Delta x)$ . The minimum value of vortex distance in Fig. 1b is 0.14 fm, this value is same as  $\sqrt{2}\Delta x$ . At this stage, both of the phases of  $^1\text{S}_0$  and  $^3\text{P}_2$  condensates rotate by  $2\pi$  around the center of the box, locking the phase difference at a certain value. As  $\zeta_2$  increases, the size of the triplet vortex which contains two overlapping HQVs slightly decreases, which results in an increase of overlap of singlet and triplet order parameters. As a result,  $E_{\text{Jos}}/\zeta_2$  increases with  $\tilde{\zeta}_2$  for  $1 < \tilde{\zeta}_2 \leq 2$ , where the factor of  $|\sigma(\mathbf{r})|^2|\psi_m(\mathbf{r})|^2$  in Eq. (14) is enhanced because of the larger density overlap.

#### 2. Vortex Structure

To get deeper insight into the observed behavior, we show in Figs. 2, 3, and 4 the actual spatial configurations of the HQVs and SQV, along with the corresponding distribution of  $\mathcal{E}_{\text{Jos}}(\mathbf{r})$ , at the values of  $\tilde{\zeta}_2$  indicated by the vertical black dotted lines in Figs. 1(a) and 1(b).

Figure 2 shows the structure of the quantized superfluid vortices for  $\tilde{\zeta}_1 = 0$  and  $\tilde{\zeta}_2 = 0.01$ . Figure 2a shows the density profile  $|\Psi_{20}(\mathbf{r})|^2$  of the  $^3\text{P}_2$  order parameter, and Fig. 2b shows its phase  $\arg[\Psi_{20}(\mathbf{r})]$ . Figure 2c shows the density pro-

file  $|\sigma(\mathbf{r})|^2$  of the  $^1S_0$  order parameter, and Fig. 2d shows its phase  $\arg[\sigma(\mathbf{r})]$ . Figure 2e shows the Josephson energy density divided by  $\zeta_2$ , *i.e.*,  $\mathcal{E}_{\text{Jos}}(\mathbf{r})/\zeta_2$ , which reflects the relative phase between the  $^1S_0$  and  $^3P_2$  condensates. A single SQV of the  $^1S_0$  condensate is located at the center, identified as the point where the density vanishes (the dark region at the center) in Fig. 2c and around which the phase winds by  $2\pi$  as shown in Fig. 2d. The two  $^3P_2$  HQVs, identified analogously from the density-depleted points in Fig. 2a and the surrounding  $2\pi$  phase winding in Fig. 2b, are located point symmetrically with respect to the SQV.

In Fig. 2e, we show the normalized Josephson energy density,  $\mathcal{E}_{\text{Jos}}(\mathbf{r})/\zeta_2$ , to investigate the effects of the Josephson coupling. In the figure, the red colored region corresponds to  $\cos[2\Delta\theta_m(\mathbf{r})] > 0$  and  $\Delta\theta_m \approx 0$ , while the blue colored region corresponds to  $\cos[2\Delta\theta_m(\mathbf{r})] < 0$  and  $\Delta\theta_m \approx \pi/2$ . Since the phase of  $\sigma(\mathbf{r})$  rotates by  $2\pi$  around the center and that of  $\Psi_{20}(\mathbf{r})$  rotates around each of the two HQVs, there appear some domains with non-zero phase difference between  $^1S_0$  and  $^3P_2$  order parameters. Indeed, a pronounced variation and sign changes of the Josephson energy are observed in the vicinity of the vortex cores, where the phase fields  $\theta_\sigma(\mathbf{r})$  and  $\theta_m(\mathbf{r})$  wind rapidly. As a result, we find blue colored regions in Fig. 2e between the  $^1S_0$  SQV and the two  $^3P_2$  HQVs. This blue colored regions between a  $^1S_0$  SQV and a  $^3P_2$  HQV can be regarded as a domain wall, across which the relative phase  $\Delta\theta_m(\mathbf{r})$  changes abruptly from 0 to  $\pi/2$ . In the present case, there are two domain walls between the SQV and the two HQVs. This spatial structure reflects the relative phase locking ( $\Delta\theta_m \approx 0$ ) in the outer region and unlocking ( $\Delta\theta_m \neq 0$ ) in between SQV and HQVs. The sign change of  $\mathcal{E}_{\text{Jos}}(\mathbf{r})$  across the vortex core indicates a local reversal of the effective Josephson coupling. As can be inferred from Eq. (14), having the blue colored domains is energetically unfavorable, which we identify as a key driver of the inter-vortex force between the  $^1S_0$  and  $^3P_2$  quantized vortices.

To show such interaction dependence of vortex configurations, we show vortex configurations obtained for different Josephson coupling strength,  $\tilde{\zeta}_2 = 0.15$  and  $\tilde{\zeta}_2 = 1.00$ , in Figs. 3a–3e and Figs. 4a–4e, respectively. The meaning of each panel is the same as Fig. 2. The results shown in Figs. 3a–3d and Figs. 4a–4d very similar to those in Figs. 2a–2d, except slight decrease of vortex distance between two HQVs. In the case of  $\tilde{\zeta}_2 = 0.01$  (Fig. 2e), the sign-reversed regions of the Josephson energy density (the blue colored regions) have a finite width and length. As  $\tilde{\zeta}_2$  increases to 0.15 (Fig. 3e), this region becomes narrower and shorter, and for  $\tilde{\zeta}_2 = 1.00$  (Fig. 4e), it has essentially disappeared. Correspondingly, the distance between the two HQVs also decreases with increasing  $\tilde{\zeta}_2$ : at  $\tilde{\zeta}_2 = 0.15$  (Fig. 3a) the separation is roughly half of that at  $\tilde{\zeta}_2 = 0.01$  (Fig. 2a), and at  $\tilde{\zeta}_2 = 1.00$  (Fig. 4a) the two HQVs are nearly adjacent to each other. This result demonstrates that the Josephson coupling  $\zeta_2$  acts as a tunable parameter controlling the effective Josephson-like force between the  $^1S_0$  SQV and the  $^3P_2$  HQVs, with direct implications for the vortex depinning dynamics relevant to the glitch mechanism. This alignment indicates

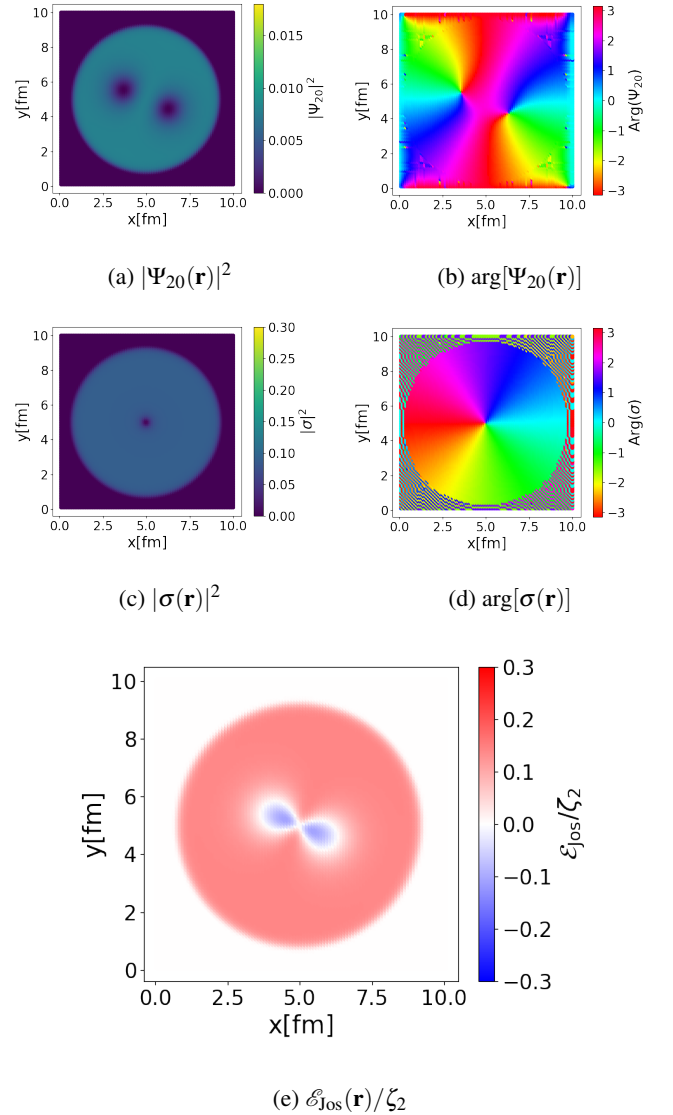


FIG. 2: Color maps of the equilibrium configuration obtained by imaginary-time evolution without pinning potentials for both condensates and  $\zeta_1 = 0$ ,  $\tilde{\zeta}_2 = 0.01$ . (a) Absolute value of the  $^3P_2$  condensate,  $|\Psi_{20}(\mathbf{r})|^2$ , showing the two HQV cores as density-depleted regions. (b) Phase of the  $^3P_2$  condensate,  $\arg[\Psi_{20}(\mathbf{r})]$ , displaying the  $2\pi$  phase winding around each HQV core. (c) Absolute value of the  $^1S_0$  condensate,  $|\sigma(\mathbf{r})|^2$  showing the SQV core as a single density-depleted region. (d) Phase of the  $^1S_0$  condensate,  $\arg[\sigma(\mathbf{r})]$  displaying the  $2\pi$  phase winding around the SQV core. (e) Spatial distribution of the Josephson energy density,  $\mathcal{E}_{\text{Jos}}(\mathbf{r})/\zeta_2$ , reflecting the relative phase between the two condensates. All panels share the same spatial coordinates  $(x, y)$ .

that a configuration in which the two HQVs and one SQV are co-located is energetically more favorable than one in which the HQVs are spatially separated from the SQV, since the former minimizes the total Josephson energy. As a result, the

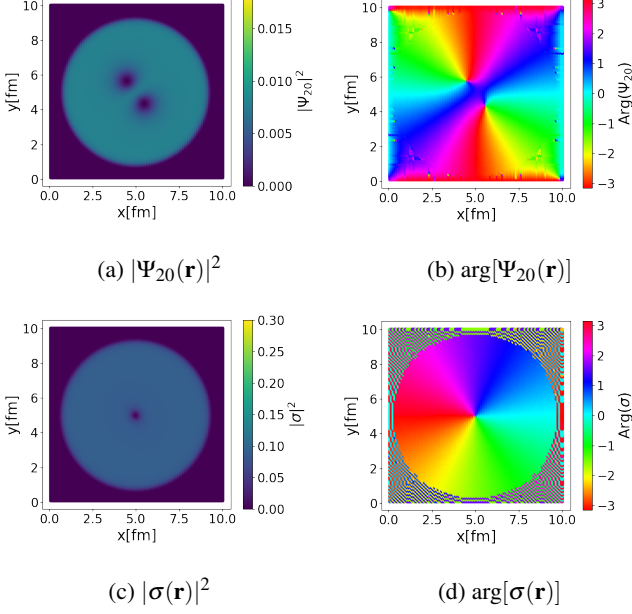


FIG. 3: Same as Fig. 2, but for  $\zeta_1 = 0$  and  $\tilde{\zeta}_2 = 0.15$ .

two HQVs and the SQV interact attractively, and the system evolves toward a bound vortex configuration with increasing  $\tilde{\zeta}_2$ .

### B. Simulation with Pinning Site

In the previous section, it is found that the Josephson coupling induces the effective attractive interaction between SQV and HQVs that collocate them at the same position. Such attractive interaction may significantly affect the vortex pinning mechanism near the crust-core boundary. To investigate how the Josephson coupling affects pinning configurations, here we introduce artificial pinning potentials for  $^1\text{S}_0$  SQV and  $^3\text{P}_2$  HQVs. We set a pinning potential, defined in Sec. III B, for SQV at the center of the container, while two pinning poten-

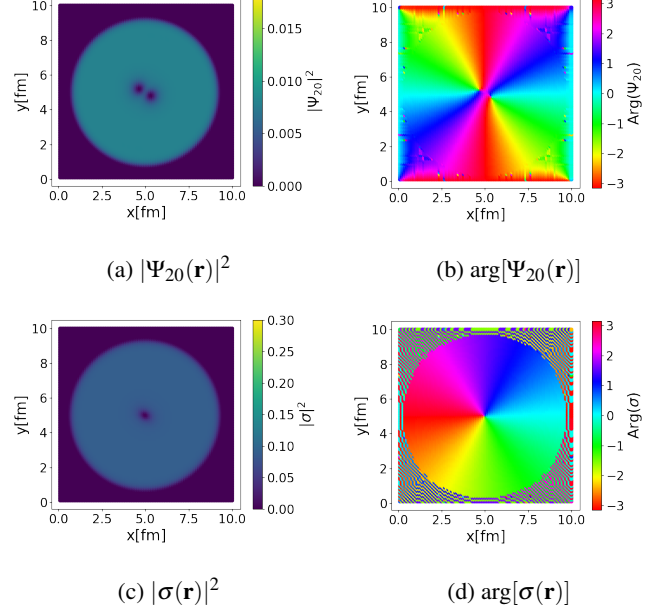


FIG. 4: Same as Figs. 2 and 3, but for  $\zeta_1 = 0$  and  $\tilde{\zeta}_2 = 1$ .

tials are located at  $(\pm \frac{d_{\text{pin}}}{2}, 0)$  with  $d_{\text{pin}} = 2, 3$ , and  $4$  fm. We perform calculations for various values of the Josephson coupling constant,  $0 \leq \tilde{\zeta}_2 \leq 2$ , keeping  $\zeta_1 = 0$  for simplicity, to analyze when vortices unpinning from the pinning site, and check the vortices structures during unpinning.

#### 1. Energy and Vortex Distance

Figure 5a shows the  $\tilde{\zeta}_2$ -dependence of the Josephson term energy normalized by  $\tilde{\zeta}_2$ ,  $E_{\text{Jos}}/\tilde{\zeta}_2$ , for three different pinning-site separations between the two HQVs:  $2.0$  fm (red),  $3.0$  fm (blue), and  $4.0$  fm (green). Figure 5b shows the corresponding change in the actual inter-vortex distance of the two HQVs as  $\tilde{\zeta}_2$  is varied, using the same color scheme as in Fig. 5a.

As seen in Fig. 5a, the Josephson energy term behaves sim-

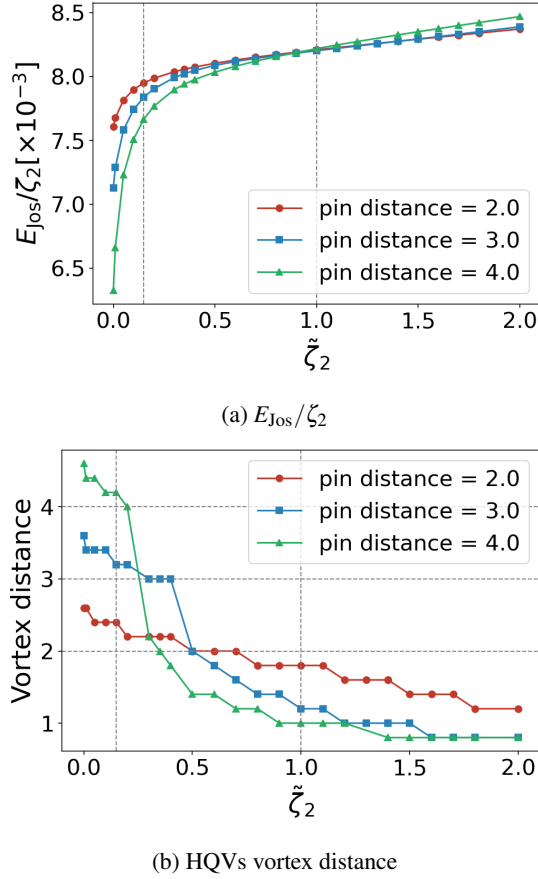


FIG. 5: (a) Josephson energy,  $E_{\text{Jos}}(\mathbf{r})/\zeta_2$ , as a function of the Josephson coupling constant  $\tilde{\zeta}_2$  for three values of the pinning site distances of HQV, as indicated in the legend. Here,  $\zeta_1 = 0$ . The system consists of a single SQV in the  $^1S_0$  condensate and two HQVs in the  $^3P_2$  condensate, with singlet and triplet pinning position. (b) Inter-vortex distance between the two HQVs in the  $^3P_2$  condensate as a function of the Josephson coupling constant  $\tilde{\zeta}_2$  for three values of the pinning site of HQV, as indicated in the legend. The system configuration and imaginary-time evolution procedure are the same as in Fig. 5a. In both panels, the results for HQV distance of 2.0, 3.0, and 4.0 fm are shown by red filled circles connected with solid lines, blue filled squares connected with solid lines, and green filled triangles connected with solid lines, respectively.

ilarly regardless of the pinning-site separation: it increases sharply with  $\tilde{\zeta}_2$ , and then grows nearly linearly beyond  $\tilde{\zeta}_2 \simeq 0.25$ . We note that the magnitude of  $E_{\text{Jos}}/\zeta$  can not directly be compared with that shown in Fig. 1b, because the density distribution is substantially modified due to the presence of three pinning potentials. Namely, the pinning potentials expel the densities and the condensate densities in the outer region are kept nearly constant even when vortices are unpinned with increasing  $\tilde{\zeta}_2$ , which results in the monotonic increase of  $E_{\text{Jos}}/\zeta_2$  in Fig. 5b.

As shown in Fig. 5b, when  $\zeta_2$  is increased beyond a criti-

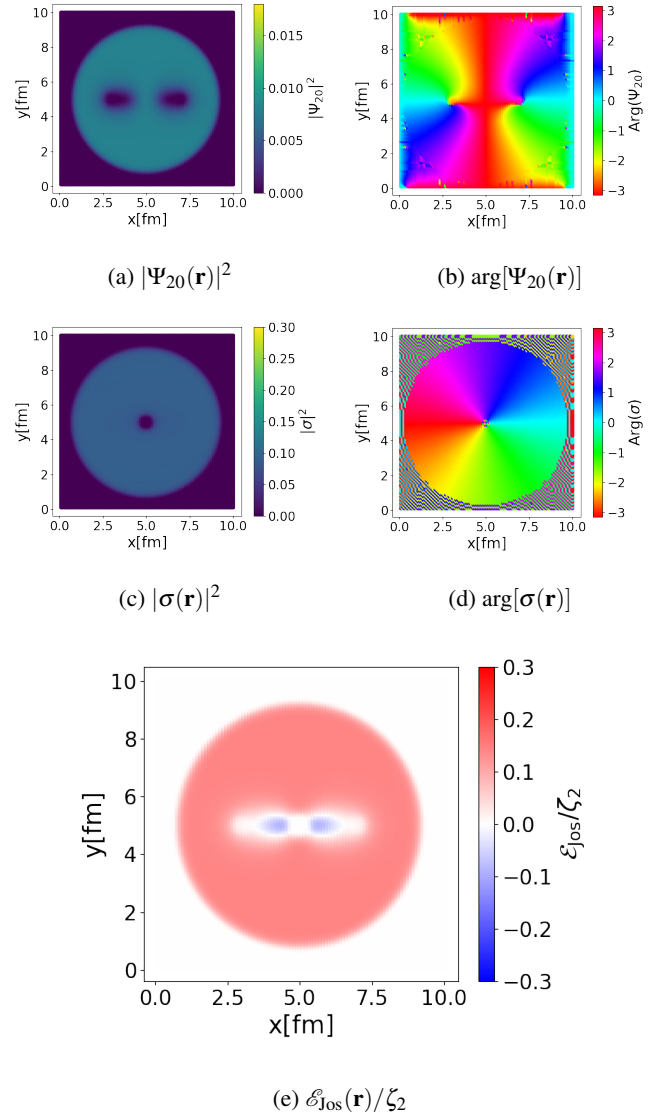


FIG. 6: Same as Figs. 2, 3, and 4, but for  $\zeta_1 = 0$  and  $\tilde{\zeta}_2 = 0.15$ , with pinning potentials, where HQV pinning site distance is  $d_{\text{pin}} = 4.0$  fm.

cal value, the HQVs are no longer confined to their pinning sites and are instead attracted toward the SQV. The critical value depends on the pinning-site separation which is found to be:  $\tilde{\zeta}_2 = 0.20$  and  $0.50$  for a pinning-site separation of  $d_{\text{pin}} = 4.0$  fm and  $3.0$  fm, respectively. However, for the  $d_{\text{pin}} = 2.0$  fm case, the depinning was not as abrupt, which occurs gradually around  $0.70 < \tilde{\zeta}_2 < 1.25$ , since the initial separation was already short. The vortices pinned at closer separations require a larger  $\tilde{\zeta}_2$  to unpin, indicating that the effective attractive force is weaker at shorter inter-vortex distances. After unpinning, the resulting inter-vortex distances for the  $3.0$  fm and  $4.0$  fm cases converge to similar final values. The resulting inter-vortex distance closely coincides with that obtained in the absence of pinning potentials (blue squares in

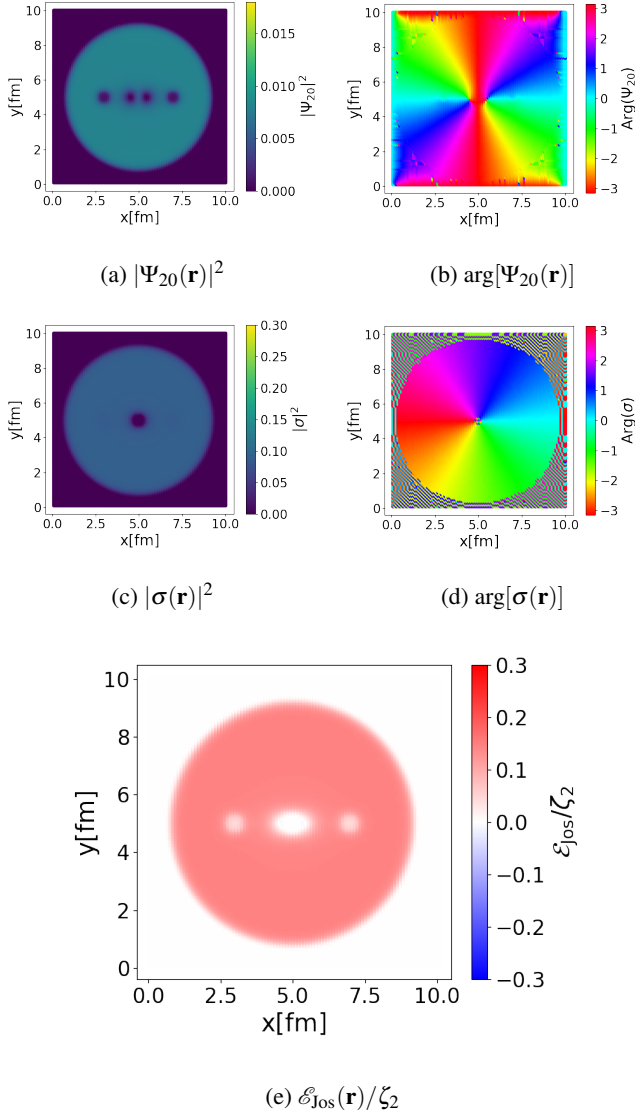


FIG. 7: Same as Fig. 6, but for  $\zeta_1 = 0$  and  $\tilde{\zeta}_2 = 1$ .

Fig. 1b), confirming that the Josephson-energy-driven attraction dominates over the pinning force for sufficiently large  $\zeta_2$ .

## 2. Vortex Structure

The vortex configurations before and after unpinning, for a pinning-site separation of  $d_{\text{pin}} = 4.0$  fm, are shown in Fig. 6 (before unpinning) and Fig. 7 (after unpinning). Each panel of Figs. 6 and 7 shows the same quantity as shown in Figs. 2, 3, and 4.

From Fig. 6e, we find that, in the weak-coupling regime, where  $\zeta_2$  is small and the vortices remain pinned, the domain walls—the region where the sign of the Josephson energy density is reversed (the blue region in Fig. 6e)—corresponding to where the phases of the  $^3\text{P}_2$  and  $^1\text{S}_0$  condensates are out of

phase—are distributed narrowly, so as to minimize its area. As a result, the phase of  $\Psi_{20}(\mathbf{r})$  around the  $^3\text{P}_2$  vortices is significantly distorted, as seen in Fig. 6b, reflecting the strong attractive force exerted on the HQVs by the Josephson term. In other words, the velocity field of  $^3\text{P}_2$  superfluid is largely distorted because of the Josephson coupling.

Once unpinning eventually occurs, the Josephson energy density becomes positive throughout the entire domain, as shown in Fig. 7e, indicating that the phases of the  $^3\text{P}_2$  and  $^1\text{S}_0$  condensates become aligned. Consequently, the two HQVs bind to the SQV after unpinning, forming a composite vortex structure (*cf.* Figs. 7a and 7c). The resulting inter-vortex distance in this case is nearly identical to that obtained in the absence of pinning sites.

## V. SUMMARY AND PROSPECT

In this work, we have investigated the microscopic interaction between half-quantized vortices (HQVs) in the  $^3\text{P}_2$  neutron superfluid and a singly-quantized vortex (SQV) in the  $^1\text{S}_0$  neutron superfluid in a two-dimensional coexistence phase, based on the Gross-Pitaevskii (GP) framework. We have demonstrated that the Josephson coupling which gives rise to the Josephson energy term [Eq. (14)], induces a significant attractive interaction between the two  $^3\text{P}_2$  HQVs and one  $^1\text{S}_0$  SQV. This attraction is strong enough to overcome the pinning potential applied to the  $^3\text{P}_2$  vortices, driving them away from their pinning sites and toward the SQV. The resulting HQV–SQV–HQV bound configuration substantially alters the spatial arrangement of the  $^3\text{P}_2$  vortices, providing microscopic evidence that the boojum structure proposed in Ref. [84] can indeed be formed at the phase boundary between the inner crust and the outer core of a neutron star. Furthermore, the strong attraction exerted on the  $^1\text{S}_0$  SQV by the  $^3\text{P}_2$  HQVs suggests that, if the triplet vortices are pinned by some mechanism, the singlet vortex will also experience an attractive force from the triplet vortices, potentially suppressing the relative motion of the two vortex species and affecting the dynamics relevant to the glitch mechanism.

We note that the present study constitutes a first step toward evaluating the role of the boojum structure in neutron star glitches, *e.g.* in the vortex network model of Ref. [84], microscopically. The simulations are performed in two spatial dimensions under the assumption of a coexistence phase, which requires the phase boundary to be planar. A fully three-dimensional treatment will be needed to capture the realistic geometry of the inner-crust–outer-core boundary.

The present results motivate several directions for future work. First, three-dimensional simulations will allow us to evaluate how the HQV bends in the vicinity of a boojum, and whether the HQV–SQV connection can form a large-scale vortex network of the kind envisioned in Ref. [84]. Three-dimensional calculations are already underway [119]. It is also important to investigate the effects of the Josephson coupling on vortex lattices, as studied in two-component BEC systems, since the collective behavior of coupled vortex lattices—in particular, whether the  $^1\text{S}_0$  and  $^3\text{P}_2$  vortex lat-

tices lock together or slide independently—is directly relevant to the angular momentum transfer mechanism underlying the glitch phenomenon [92]. Second, the effect of the HQV–SQV attraction on the dynamics of the SQV itself will be assessed, as the suppression of SQV motion by pinned triplet vortices may have direct implications for the avalanche model of pulsar glitches. We note here that we can naturally extend our formalism for real-time evolution to investigate complex vortices dynamics, including the mixed phase, under various conditions.

These microscopic results on the vortex–vortex coupling will then be used to evaluate the influence of the HQV–SQV attraction on the vortex network model and the avalanche mechanism, with the aim of providing a quantitative microscopic basis for the glitch mechanism. Additionally, the formation of a stable HQV–SQV–HQV molecular bound state opens the possibility of a novel oscillation mode arising from the attractive inter-vortex interaction. Such an inter-vortex vibrational mode could serve as a new internal heat source in neutron stars, with potential implications for neutron star cooling. The possibility of recombination and exchange of the HQV–SQV–HQV pair upon vortex collisions will also be examined in future work. For instance, exchanging a partner in a collision of two vortex molecules found in a two-component Bose-Einstein condensation (BEC) [96] may occur.

The significant role of the Josephson term found in this work suggests broader implications beyond neutron star physics. In finite nuclei, both spin-singlet and spin-triplet pairings are in principle possible, and recent studies have shown that spin-triplet pairing condensates can coexist with spin-singlet pairing in even-even singly closed nuclei such as Ca and Sn isotopes [120]. The present results indicate that the Josephson coupling between singlet and triplet condensates—driven by their relative phase—can generate a strong inter-condensate interaction, suggesting that analogous Josephson effects may play an important role in finite nuclei where both  $^3P_2$  and  $^1S_0$  pairings coexist, as well as in proton–neutron systems coupled via the  $^3S_1$  channel. To quantify these effects in realistic nuclear systems, it will be necessary to determine

the triplet pairing coupling constant and the singlet–triplet interaction coefficient from experimental data on heavy nuclei such as Pb isotopes.

Furthermore, recent microscopic calculations based on superfluid band theory have indicated the emergence of spin-triplet pairing in the rod (pasta) phase of the neutron star inner crust, where two-dimensional crystalline structures, spin-orbit interactions, and strong magnetic fields act in concert [121]. Since the pasta phase is expected to occupy the high-density region of the inner crust near the crust–core boundary, the vortex dynamics in the vicinity of the pasta phase may be influenced by the inter-condensate Josephson coupling identified in the present work, with potential implications for the glitch mechanism in this transitional region.

Finally, the inter-condensate interaction studied here is directly analogous to the Josephson coupling in two-component BEC systems, as recently realized experimentally in bilayer superfluid systems [122,123]. The present framework is therefore also relevant to the broader context of multi-component condensate physics, including the dynamics of coupled half-integer and integer vortices in spinor BEC systems [53,112].

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